

GENERAL SCIENCE: CHEMISTRY FUNDAMENTALS

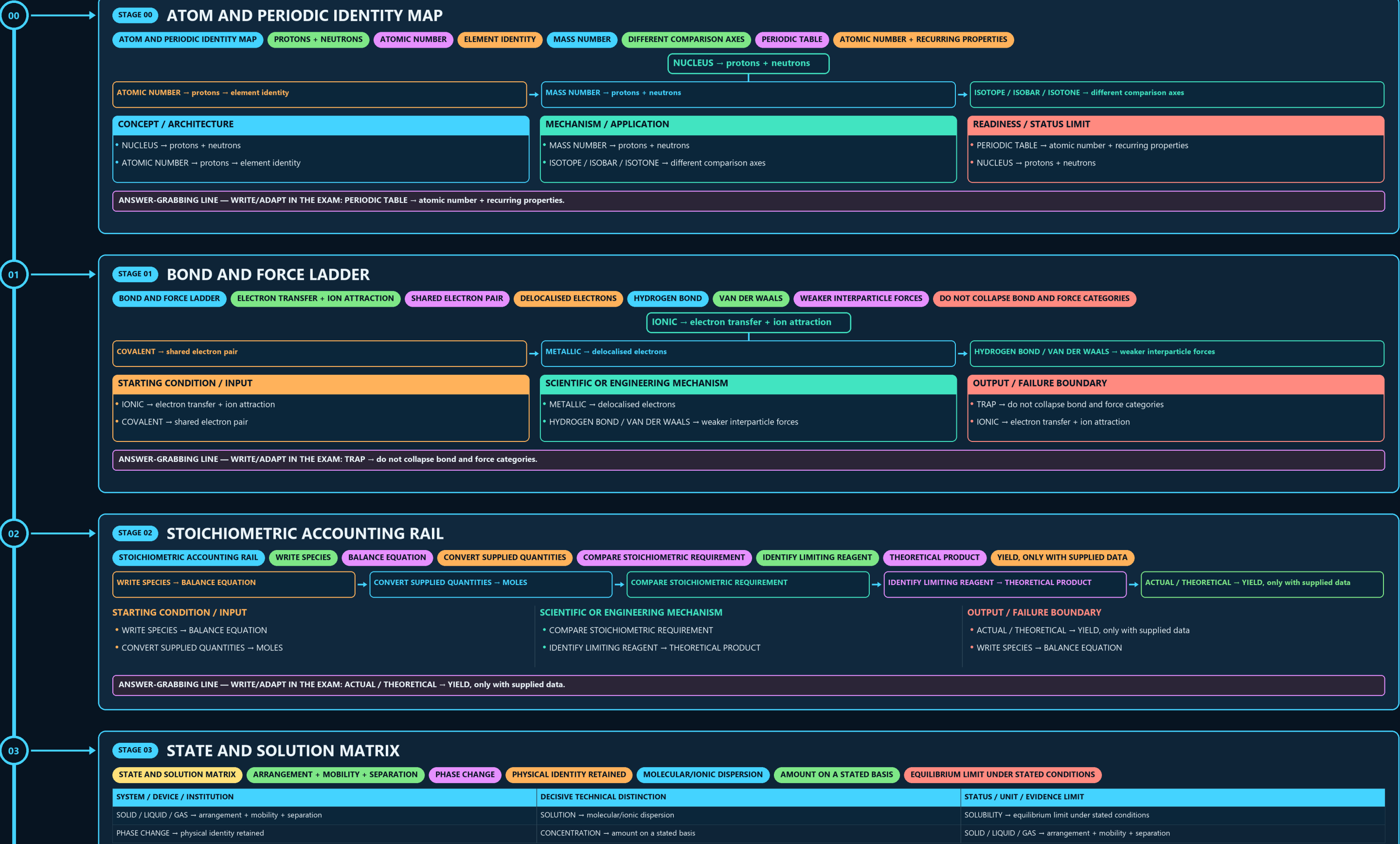
ATOM AND PERIODIC IDENTITY → BOND AND FORCE LADDER → STOICHIOMETRIC ACCOUNTING RAIL → STATE AND SOLUTION MATRIX → ACID BASE BUFFER CONTROL → ELECTROCHEMICAL TWO-CELL MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

STARTING CONDITION / INPUT

- WRITE SPECIES → BALANCE EQUATION
- CONVERT SUPPLIED QUANTITIES → MOLES

SCIENTIFIC OR ENGINEERING MECHANISM

- COMPARE STOICHIOMETRIC REQUIREMENT
- IDENTIFY LIMITING REAGENT → THEORETICAL PRODUCT

OUTPUT / FAILURE BOUNDARY

- ACTUAL / THEORETICAL → YIELD, only with supplied data
- WRITE SPECIES → BALANCE EQUATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACTUAL / THEORETICAL → YIELD, only with supplied data.

04

STAGE 03 STATE AND SOLUTION MATRIX

STATE AND SOLUTION MATRIXARRANGEMENT + MOBILITY + SEPARATIONPHASE CHANGEPHYSICAL IDENTITY RETAINEDMOLECULAR/IONIC DISPERSIONAMOUNT ON A STATED BASIS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SOLID / LIQUID / GAS → arrangement + mobility + separation	SOLUTION → molecular/ionic dispersion	SOLUBILITY → equilibrium limit under stated conditions
PHASE CHANGE → physical identity retained	CONCENTRATION → amount on a stated basis	SOLID / LIQUID / GAS → arrangement + mobility + separation

SYSTEM / DEVICE / INSTITUTION

- SOLID / LIQUID / GAS → arrangement + mobility + separation
- PHASE CHANGE → physical identity retained

DECISIVE TECHNICAL DISTINCTION

- SOLUTION → molecular/ionic dispersion
- CONCENTRATION → amount on a stated basis

STATUS / UNIT / EVIDENCE LIMIT

- SOLUBILITY → equilibrium limit under stated conditions
- SOLID / LIQUID / GAS → arrangement + mobility + separation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOLID / LIQUID / GAS → arrangement + mobility + separation.

05

STAGE 04 ACID BASE BUFFER CONTROL LOOP

ACID BASE BUFFER CONTROL LOOPACID DONATES H+ ↔ BASE ACCEPTS H PHLOGARITHMIC ACTIVITY MEASUREACIDIC/BASIC CHARACTER REDUCEDCONSUMES LIMITED ADDED ACID/BASESALT SOLUTIONNEUTRAL DEPENDS ON IONS

CONCEPT / ARCHITECTURE

- ACID DONATES H+ ↔ BASE ACCEPTS H pH → logarithmic activity measure
- NEUTRALISATION → acidic/basic character reduced

MECHANISM / APPLICATION

- BUFFER → consumes limited added acid/base
- SALT SOLUTION → acidic / basic / neutral depends on ions

READINESS / STATUS LIMIT

- ACID DONATES H+ ↔ BASE ACCEPTS H pH → logarithmic activity measure
- NEUTRALISATION → acidic/basic character reduced

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SALT SOLUTION → acidic / basic / neutral depends on ions.

06

STAGE 05 ELECTROCHEMICAL TWO-CELL MAP

ELECTROCHEMICAL TWO-CELL MAPGALVANIC CELLCHEMICAL ENERGY TO ELECTRICAL ENERGYELECTROLYTIC CELLELECTRICAL ENERGY DRIVES CHEMISTRYAPPLIED ROUTESSIGN TRAPCELL TYPE CHANGES SIGNS, NOT REACTION LOCATIONS

REDUCTION → cathode

OXIDATION → anode

GALVANIC CELL → chemical energy to electrical energy

ELECTROLYTIC CELL → electrical energy drives chemistry

STARTING CONDITION / INPUT

- OXIDATION → anode
- REDUCTION → cathode

SCIENTIFIC OR ENGINEERING MECHANISM

- GALVANIC CELL → chemical energy to electrical energy
- ELECTROLYTIC CELL → electrical energy drives chemistry

OUTPUT / FAILURE BOUNDARY

- CORROSION / PLATING / EXTRACTION → applied routes
- SIGN TRAP → cell type changes signs, not reaction locations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIGN TRAP → cell type changes signs, not reaction locations.

07

STAGE 06 KINETICS VERSUS EQUILIBRIUM

KINETICS VERSUS EQUILIBRIUMRATE AXISCOLLISIONS + ACTIVATION ENERGYALTERNATIVE LOWER-ENERGY PATHWAYEQUILIBRIUM AXISFORWARD RATE EQUALS REVERSE RATEFASTER APPROACH IN BOTH DIRECTIONSNO SHIFT

CATALYST → alternative lower-energy pathway

RATE AXIS → collisions + activation energy

EQUILIBRIUM AXIS → forward rate equals reverse rate

CATALYST → faster approach in both directions

CONCEPT / ARCHITECTURE

- RATE AXIS → collisions + activation energy
- CATALYST → alternative lower-energy pathway

MECHANISM / APPLICATION

- EQUILIBRIUM AXIS → forward rate equals reverse rate
- CATALYST → faster approach in both directions

READINESS / STATUS LIMIT

- NO SHIFT → equilibrium position unchanged
- RATE AXIS → collisions + activation energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIGN 1KAP → cell type changes signs, not reaction locations.

06

STAGE 06 KINETICS VERSUS EQUILIBRIUM

KINETICS VERSUS EQUILIBRIUM

RATE AXIS

COLLISIONS + ACTIVATION ENERGY

ALTERNATIVE LOWER-ENERGY PATHWAY

EQUILIBRIUM AXIS

FORWARD RATE EQUALS REVERSE RATE

FASTER APPROACH IN BOTH DIRECTIONS

NO SHIFT

RATE AXIS → collisions + activation energy

CATALYST → alternative lower-energy pathway

EQUILIBRIUM AXIS → forward rate equals reverse rate

CATALYST → faster approach in both directions

CONCEPT / ARCHITECTURE

- RATE AXIS → collisions + activation energy
- CATALYST → alternative lower-energy pathway

MECHANISM / APPLICATION

- EQUILIBRIUM AXIS → forward rate equals reverse rate
- CATALYST → faster approach in both directions

READINESS / STATUS LIMIT

- NO SHIFT → equilibrium position unchanged
- RATE AXIS → collisions + activation energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EQUILIBRIUM AXIS → forward rate equals reverse rate.

07

STAGE 07 ORGANIC AND POLYMER FAMILY TREE

ORGANIC AND POLYMER FAMILY TREE

FUNCTIONAL GROUPS

ADDITION POLYMER

NO SMALL-MOLECULE ELIMINATION

CONDENSATION POLYMER

SMALL MOLECULE ELIMINATED

THERMOPLASTIC ↔ THERMOSET

REHEATING BOUNDARY

FUNCTIONAL GROUPS → alcohol

ADDITION POLYMER → no small-molecule elimination

CONDENSATION POLYMER → small molecule eliminated

THERMOPLASTIC ↔ THERMOSET → reheating boundary

CONCEPT / ARCHITECTURE

- FUNCTIONAL GROUPS → alcohol
- ADDITION POLYMER → no small-molecule elimination

MECHANISM / APPLICATION

- CONDENSATION POLYMER → small molecule eliminated
- THERMOPLASTIC ↔ THERMOSET → reheating boundary

READINESS / STATUS LIMIT

- HYDROGEL → cross-linked hydrophilic network
- FUNCTIONAL GROUPS → alcohol

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FUNCTIONAL GROUPS → alcohol | aldehyde | ketone | acid | amine | ester.

08

STAGE 08 MATERIALS STRUCTURE-PROPERTY WHEEL

MATERIALS STRUCTURE-PROPERTY WHEEL

TETRAHEDRAL NETWORK

LAYERED STRUCTURE

DISTINCT CARBON ARCHITECTURES

COMPOSITION + PROCESSING ALTER PROPERTIES

SAME ELEMENT CAN YIELD DIFFERENT MATERIAL BEHAVIOUR

CONCEPT / ARCHITECTURE

- DIAMOND → tetrahedral network
- GRAPHITE → layered structure

MECHANISM / APPLICATION

- GRAPHENE / FULLERENE / NANOTUBE → distinct carbon architectures
- ALLOY → composition + processing alter properties

READINESS / STATUS LIMIT

- RULE → same element can yield different material behaviour
- DIAMOND → tetrahedral network

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → same element can yield different material behaviour.

09

STAGE 09 FUEL CONVERSION FAULT TREE

FUEL CONVERSION FAULT TREE

OXIDATION + USEFUL HEAT + POSSIBLE POLLUTANTS

INCOMPLETE COMBUSTION

DIFFERENT PRODUCT BURDEN

CONTROLLED OXYGEN/STEAM CONVERSION

CARBON MONOXIDE + HYDROGEN DOMINANT ROUTE

OUTCOME FIREWALL

PRODUCT IS NOT EFFICIENCY OR CLIMATE PROOF

COMBUSTION → oxidation + useful heat + possible pollutants

INCOMPLETE COMBUSTION → different product burden

GASIFICATION → controlled oxygen/steam conversion

SYNGAS → carbon monoxide + hydrogen dominant route

CONCEPT / ARCHITECTURE

- COMBUSTION → oxidation + useful heat + possible pollutants
- INCOMPLETE COMBUSTION → different product burden

MECHANISM / APPLICATION

- GASIFICATION → controlled oxygen/steam conversion
- SYNGAS → carbon monoxide + hydrogen dominant route

READINESS / STATUS LIMIT

- OUTCOME FIREWALL → product is not efficiency or climate proof
- COMBUSTION → oxidation + useful heat + possible pollutants

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME FIREWALL → product is not efficiency or climate proof.

